

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. (CE, IT, EC) Examination

Dec 2012

EE102 Fundamentals of Electrical Engineering

Date: 10.12.2012, Monday

Time: 01:30 p.m. To 04:30 p.m.

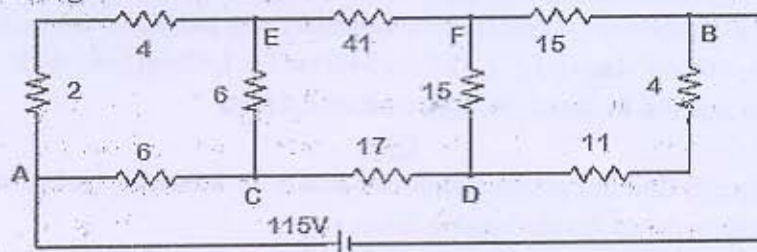
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Derive the equation $V_c = V [1 - e^{-t/RC}]$ for charging of a capacitor. [04]
(b) Two capacitors having capacitance of $6\mu F$ and $10\mu F$ are connected in parallel. A $16\mu F$ capacitor is connected in series with this combination and complete circuit is connected across 400 Volts. Calculate: (i) Total Capacitance of circuit, (ii) Total charge in the circuit, (iii) Voltage across each capacitor. [03]
- Q - 2 (a) Explain effect of increase of temperature on resistance in case of (i) Metals, (ii) Insulators, (iii) Alloys. [03]
(b) Using Star-Delta transformation principle, determine the total Current taken by circuit shown in the Figure and also find current in the 17Ω resistor in the network. [07]



- (c) Four resistances $R_1 = 56\Omega$, $R_2 = 50\Omega$, $R_3 = 40\Omega$ & R_4 are in parallel across 280V DC supply. Current in 56Ω is 5A (i.e. $I_1 = 5A$) and total current $I = 37.6A$. Draw the circuit diagram as per description and find the values of I_2 , I_3 , I_4 and resistance of R_4 . [04]
- OR
- Q - 2 (a) Give the comparison between Electrical circuits and Magnetic circuits. [07]
(b) Define the following terms. (Any three) [03]
(i) Leakage coefficient, (ii) Dielectric constant, (iii) Electric flux density,
(iv) Relative Permeability.
(c) An iron ring having cross sectional area of 6 cm^2 is wound with a wire of 100 turns and has a saw cut of 2 mm. Calculate the magnetizing current required to produce a flux of 0.1 mwb if the mean length of the magnetic path is 30 cm and relative permeability of iron is 470. [04]
- Q - 3 (a) Explain why the electrical appliances are connected in parallel but not in series and give the comparison of series and parallel circuit. [04]
(b) State and explain Kirchhoff's laws applied to electric circuit. [04]
(c) An iron ring having cross sectional area of 20 cm^2 and a mean diameter of 18cm has a coil of 270 turns uniformly wound over it. A current of 1.27A flows through the coil [06]

which produces a flux 1.13mwb in the ring. Find the reluctance of the circuit, the absolute and relative permeabilities of iron.

OR

- Q - 3 (a) Derive the voltage divider equation for series and current divider equation for parallel connections of resistors. [04]
- (b) Capacitor of 50 microfarad is connected in series with 100 ohm resistor is suddenly connected across 100 V DC supply. Calculate (i) Time constant of the circuit, (ii) Initial current, (iii) Voltage across resistor after 6 mili sec. [03]
- (c) A parallel plate capacitor has a plate area of 4cm^2 . The plates are separated by three slabs of different dielectric materials of thickness 0.3, 0.4 and 0.3mm with the relative permittivities of 3, 2.5 and 2 respectively. Calculate the capacitance of each material if the supply is of 100V. [07]

SECTION - II

- Q - 4 (a) Define the terms with respect to three phase AC system. [03]
- (i) Phase sequence, (ii) Balanced load, (iii) Phase and line voltage.
- (b) The two wattmeter method is used to measure the power of balanced 3 phase AC system having inductive load. Two wattmeters' readings are 2500W and 450W respectively. Find the power factor when (i) Both readings are positive, [04]
- (ii) Only the second reading is negative.
- Q - 5 (a) Explain the principle for the generation of alternating voltage and current. Also obtain the equations of the alternating voltage and current for single phase. [08]
- (b) The values of instantaneous currents in the branches of a parallel circuit are as follow: $i_1=5\sin 346t$; $i_2=10\sin(346t+\pi/4)$; $i_3=7.5\sin(346t+\pi/2)$; $i_4=8\sin(346t-\pi/3)$; Determine the RMS value and the frequency of the resultant current. [06]

OR

- Q - 5 (a) For 3 phase circuit, find out power and power factor for balanced system with the use of two wattmeter method for the lagging loads. [08]
- (b) Two impedances $Z_1 = 6+j8 \Omega$ and $Z_2 = 8-j6 \Omega$ are connected in parallel across AC supply source of 100V. Find (i) Current & power factor of each branch, (ii) Overall current & power factor of the circuit, (iii) Power consumed by each impedance. [06]
- Q - 6 (a) An inductive coil of resistance R and inductance L is connected in series with a capacitor of C. Derive the expression of bandwidth. [08]
- (b) An inductive coil of resistance 32Ω and reactance 15.7Ω is connected in series with a capacitor of reactance 79.5Ω the circuit as a whole is connected across a 500V AC supply. Determine: (i) Current, (ii) Angle of phase difference between the applied voltage and current, (iii) Total power absorbed. [06]

OR

- Q - 6 (a) Give the comparison of series and parallel resonance. [07]
- (b) An RLC series circuit with resistance of 10Ω , inductance of 0.2H and capacitance of $40\mu\text{F}$ is supplied by a 100V AC supply at variable frequency. Find the following with respect to series resonant circuit: (1) The frequency at which the resonance takes place, (2) Current, (3) Power factor, (4) Voltage across R, L, and C at that time, (5) Quality factor, (6) Half power points, (7) Phasor diagram. [07]
